Lab 40 Worksheet — Malware Types, Symptoms and the Seven-Step Removal Procedure

Name: ____ **Date:** __

Exam objective: Identify malware types by behaviour and apply the CompTIA seven-step malware removal procedure in order (Core 2 objectives 2.3 and 3.3).

Record as you go

Step	What you did	What you observed
1	Open https://alfredang.github.io/cybersecuritysimulator/ and select the Malware module, then run the visual demonstration. Nothing real executes — the animation is conceptual.	
2	Run the Virus Spread Visualizer and record how propagation accelerates, then state why one infected machine on a flat network is an organisational problem rather than a user problem.	
3	Open the Ransomware module and use the locked screen demonstration to see what a victim actually sees, including the countdown and the payment demand.	
4	Record the ransomware response: do not pay, isolate the machine immediately, identify the variant, and restore from a backup that is offline or air-gapped.	
5	Explain why an air-gapped or immutable backup is the only reliable ransomware defence: ransomware encrypts every backup it can reach over the network.	
6	Build the malware reference table with rows for virus, worm, trojan, ransomware, rootkit,	

Step	What you did	What you observed
	spyware, keylogger, cryptominer and botnet.	
7	For each type record the propagation method, the primary symptom and the specific removal difficulty, and note that a rootkit may require reinstalling the operating system.	
8	Record the distinguishing behaviours precisely: a virus needs a host file and human action, while a worm self-propagates across the network with no user involvement at all.	
9	Write the seven-step removal procedure in order: investigate and verify symptoms; quarantine the system; disable System Restore; remediate by updating the anti-malware and scanning in safe mode; schedule scans and updates; re-enable System Restore and create a restore point; educate the end user.	
10	Explain why System Restore must be disabled at step three: restore points can hold the malware, so remediating before disabling them leaves a re-infection path.	
11	Explain why safe mode matters at step four: it loads a minimal driver set, so malware that hooks at startup is not running to defend itself.	
12	Apply the full procedure to a scenario — a user reports pop-ups, a slow machine and a browser homepage they did not set — writing the specific action you take at each of the seven steps.	

Verification

Success criterion: Your reference table covers at least eight malware types with propagation and symptoms; the seven steps are in the correct order; you correctly explain why System Restore is disabled before remediation; and the scenario has a specific action at every step.

- I completed every step in the lab.
- My result meets the success criterion above.

- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
